

Module 01: Systems — Your Amazing Body

Learning Objectives

1. Understand that a **Body** is a System of systems (Digestion, Sleep, Senses).
2. Learn how the different parts work together (Stomach needs Teeth!).
3. Know that growing requires energy and care.

Introduction: The Living Machine

Your body is the most amazing machine in the world. It can heal cuts. It can turn an apple into energy. It can dream! But like any machine, it has parts that must work together.

- **The Engine:** Heart and Lungs.
- **The Computer:** Brain.
- **The Wheels:** Legs and Feet.

Key Concepts

1. Inputs and Outputs

A system needs inputs.

- **Input:** Food, Water, Air, Hugs. A system makes outputs.
- **Output:** Energy, Poop, Words, Smiles. If you put junk in (Bad Food), the machine doesn't run well!

2. Homeostasis (The Thermostat)

Your body likes to be "Just Right."

- Too Hot? -> You sweat (Cool down).
- Too Cold? -> You shiver (Warm up). This is your body's way of keeping the system safe.

3. Growth (The Upgrade)

Unlike a car, your body upgrades itself!

- Bones get longer.
- Muscles get stronger.
- Brain gets smarter. Sleep is when the upgrades happen. That's why babies sleep so much!

Activities

Activity 1: The Heart Beat Check

Find your pulse (Heartbeat).

- Sit still. Count for 10 seconds. (Slow).
- Jump for 1 minute. Count again. (Fast!). Why did it speed up? The "Engine" had to work harder to deliver oxygen to the "Wheels" (Muscles). The system adapts!

Activity 2: Draw Your Insides

Lie on a big piece of paper. Trace your body outline. Draw the systems inside:

- Red tubes for Blood.
- Blue bag for Stomach.
- Squiggly lines for Brain.

Summary

You are a system of systems. By eating good food, sleeping well, and listening to your body signals, you keep your amazing machine running perfectly.

References

- *Me and My Amazing Body* by Joan Sweeney
- *Inside Your Outside* by Tish Rabe (Cat in the Hat)